A Tax on Luxury Consumption

Adrien Fabre* Matthias Kalkuhl† Lennart Stern‡ Xuan Xie§

This appendix proposes a broad luxury tax as a new own EU resource, bundled with the abolition of VAT refunds for non-EU that would also generate additional revenue for member states, mostly from personal (portable) luxury goods. Both components are motivated by the finding that raising revenue via luxury taxation generates a significantly lower marginal economic burden than other taxes. Indeed, luxury taxation is justified on the grounds of status externalities and progressivity, and it taxes the wealthiest without deterring investments nor philanthropy. However, national luxury taxation is currently distorted downward due to international tax competition. In the case of portable luxury goods there is tax competition for internationally mobile shoppers and in the case of residence-based luxury goods (e.g. cars) there is tax competition for high-net-worth residents. We therefore propose to implement a new own EU resource as a luxury excise on values exceeding category-specific per-item price thresholds. We propose a uniform ad valorem rate across categories as a focal point to facilitate negotiations. Economic modeling suggests that an EU-wide additive luxury tax rate of 20% is justified on the grounds of status externalities and signaling alone, while socially optimal tax rates are likely substantially higher once progressivity benefits are taken into account. A 20% rate is estimated to raise EUR 42 billion per year for the EU budget. While a luxury tax restricted to any particular product category would create uneven relative burdens across member states, our broad product scope (covering yachts, luxury vehicles, personal luxury items, hospitality, and fine spirits) achieves a more balanced burden relative to GDP. Countries collecting higher revenue shares (e.g., France, Italy, Spain) would benefit disproportionately from the concurrent abolition of VAT refunds, which is estimated to raise up to EUR 4 billion EU-wide. This combined reform is feasible, highly progressive, and politically viable.

1 Introduction

There are strong economic reasons (reviewed below) for why taxing luxury goods at a higher rate than other goods will raise social welfare. However, if countries decide individually on their luxury good taxes then there are strong tax competition forces pushing luxury good taxation down. For portable luxury goods, countries have an incentive to lower luxury good taxes to attract and retain shoppers, thereby gaining other indirect tax revenue from their non-refundable spending on on-site consumption of other goods and services. For non-portable luxury goods that are taxed based on residence (e.g. luxury cars), there is an incentive for countries to lower the tax rate in order to attract more (internationally mobile) high earners and high wealth individuals. In the case of the post-Brexit abolition of VAT refunds in the UK, these spillover

*CNRS; CIRED; Global Redistribution Advocates.
†Potsdam Institute for Climate Impact Research (PIK)
‡PIK. Corresponding author: lennart.stern@pik-potsdam.de.
§Potsdam Institute for Climate Impact Research (PIK).

effects became very visible for the case of portable luxury goods, whose sales shifted substantially to other countries, including continental Europe [1].

This suggests that luxury good taxes are promising as an EU own resource, given that such a common tax could correct a distortion created by tax competition. Even if individual EU countries find it optimal for themselves to not introduce luxury good taxes, they might still find it in their interest to vote in favor of an EU-wide luxury tax, given that this will prevent them from losing shoppers or residents to other EU countries. Put differently, each EU country benefits from other EU countries' luxury good taxation via the resulting displacement of shoppers/residents to them. Thus, voting for an EU-wide luxury tax is more attractive.

We will argue that there are advantages to the "bundled approach" of introducing an EU-wide luxury tax to fund the EU budget as compared to the "separated approach" of introducing mandatory minimal luxury taxes and increasing the EU budget via the traditional GNI-based contributions. In other words, we will argue that the two problems, "tax competition leading to undertaxation of luxury" and "unanimity constraints leading under-provision of funding for the EU" can best be addressed via a single mechanism, namely an EU-wide luxury tax to fund the EU budget.

There are two core arguments in favor of the bundled approach. Firstly, the bundled approach avoids, in contrast to the separated approach, aggravating tax competition for people with high income and wealth. Under the separated approach, attracting such people would be made more valuable for each country to the extent that the mandatory minimal luxury taxes increase the tax revenues that such people yield. Thus countries would have increased unilateral incentives to further lower taxes on high income and wealth. Under the bundled approach, no such aggravation of (income and wealth) tax competition would result. The reason for this is that each country would be indifferent about which would collect the EU-wide luxury tax, given that all the revenue would go to the EU budget anyway.

The first argument in favor of the bundling approach that we just sketched shows that the aggregate welfare gains achievable via the bundled approach are greater than under the separated approach. This is per se desirable and it would also make it easier to get unanimity. We now sketch a second argument suggesting that even if we neglect this welfare gain via avoided aggravation of tax competition, the bundled approach might be more likely to get unanimity among EU member states.

This second argument in favor of bundling can be stated very generically: suppose we compare having votes on two reforms separately or on the combination of the two. If each of the separate reforms garner unanimity then so should the combination. However, the converse does not hold.

2 The proposals

2.1 The EU luxury surcharge

We propose a single EU-wide indirect tax, levied at a uniform ad valorem rate on the part of an item's price that exceeds a category-specific threshold, assessed on the first supply to a final consumer or on importation, and collected by VAT-registered vendors through the existing VAT return. Its yield accrues to the EU budget as an own resource. We refer to it as the European luxury surcharge.

The surcharge is defined by the following elements.

[1] Of the 162,000 non-EU tourists who claimed VAT refunds exclusively in the UK in 2019, approximately a fifth were claiming rebates in EU countries by 2023. Since this captures only EU-redirection, the true extensive margin is likely larger Bloomberg & Creery (2024).

(i) Scope. A set of product categories $\mathcal{C}$ (see below).

(ii) Thresholds. Each category $c \in \mathcal{C}$ carries a threshold $\bar{p}_c$ which can be either 0 or positive. If positive, it is expressed as an amount per item before any taxes.

(iii) Rate. A single rate t^{EU} applies uniformly across all categories on the value exceeding the threshold.

(iv) Collection. The surcharge is declared and paid on the VAT return of the person liable, in a separate box, and is subject to the same registration, invoicing, record-keeping and audit obligations as value added tax.

(v) Attribution. Amounts collected are attributed to the member state of taxation and transferred to the EU budget under the Own Resources Decision.

The tax base

Table 1.1 estimates the tax base. This estimate of the tax base, on which all our revenue estimates rely, is obtained with the method explained in a methodological note and implemented in the accompanying code. The categories could be specified by Combined Nomenclature (CN) codes, which already exist and are already used at import. For some categories, the thresholds could be set to 0 because all items in the category can be viewed as luxury. For example:

CN 7113/7114/7116: jewellery, articles of precious metal, precious stones

CN 9101/9102: watches with precious-metal cases

CN 4303: furskin apparel

CN 8802: aircraft for private or business use

Many other product categories contain both goods that clearly are luxury and goods that (arguably) are not. For these categories, positive per item thresholds are essential. Examples include:

CN 8703: passenger vehicles

CN 8903: yachts and pleasure craft

CN 2204/2208: wine and spirits

CN 9701-9706: art, antiques

Services (luxury hotels, cruises, fine dining) have no CN code and would need a definition by price per night.

Table 1.1: EU-27 spending on high-end and luxury goods and services by sector (2024E/2025), net of VAT and consumption taxes

Sector	EU-27 spending (EUR bn/yr)
Luxury cars	88
Personal luxury goods	81
Hospitality (hotels, cruises)	54
Yachts	25
Fine wines and spirits	22
Gourmet food and fine dining	18
Design and furniture	11
Total, goods and services	**299**

Notes: Own calculations. Global sector values from European Cultural and Creative Industries Alliance & Bain & Company (2025) are multiplied by the share purchased in Europe (published for personal luxury goods: 30.3%, D'Arpizio et al., 2024; constructed for other sectors) and by the EU-27 share of Bain's EU perimeter (74%). For cars, we assume a price threshold of EUR 32,000, equal to the EU mean estimated based on (UNRAE, 2025) and (DAT, 2026). Memo items outside the ECCIA sector perimeter: art and antiques ≈ EUR 5 bn. The yacht figure is estimated separately as a flow of consumption. It is a territorial (destination-basis) estimate of yacht-related consumption in EU waters for yachts above 10 m, based on the Italian tax introduced in 2012.

Specific proposals for thresholds could be anchored in values already in use elsewhere and indexed to inflation. Korea's Individual Consumption Tax applies to watches and bags above roughly EUR 1,300 and to jewellery, fur and leather goods above roughly EUR 3,200. France's *taxe forfaitaire sur les objets précieux* applies to jewellery, watches, art and antiques above EUR 5,000 per item, with the threshold assessed object by object except where the objects form a coherent set. India's tax collected at source on luxury goods applies above roughly EUR 10,000, uniformly across its notified list of watches, art, collectibles, yachts, bags and footwear. For accommodation, Florence's *imposta di soggiorno* is assessed directly against the net price per guest per night, in bands whose highest begins at EUR 200.

For motor cars the comparators are more dispersed: Australia's Luxury Car Tax threshold stands at roughly EUR 46,000, rising to roughly EUR 69,000 for zero-emission vehicles; Canada's at roughly EUR 65,000; China's at roughly EUR 110,000; and Greece's at EUR 17,000. Denmark's registration tax, the closest European analogue to a graduated luxury rate on vehicles, reaches its 85 per cent bracket at roughly EUR 8,200 and its top 150 per cent bracket at roughly EUR 25,500, once its VAT-inclusive base is restated net of VAT. Canada applies a separate and much higher threshold of roughly EUR 156,000 to vessels. A simple practical approach could be to define the price threshold as the average pre-tax car price in the EU, which is about EUR 32,000. This would automatically index it to inflation.

2.2 Abolishing the VAT refunds

We propose to combine the introduction of the European luxury surcharge with an EU-wide abolition of VAT refunding. VAT refunds mostly concern luxury goods, as only for high-value goods it is worth it for consumers to carry them home.

Legality

There appears to be no papers or reports questioning the legality of abolishing VAT refunds. Consistently with that, no country opposed the UK when its abolished its VAT refund after leaving the EU.

2.3 Tax design

The case for a uniform tax rate Purely from the perspective of aggregate welfare, it might be optimal to substantially differentiate the tax rates on different luxury goods, to the extent that they differ in the degree to which their consumption is motivated by status and the degree to which their consumption is correlated with income and wealth. However, a proposal for a new EU own resource might be more easily adopted in a simple form with a uniform rate. The burden of taxes on luxury cars will disproportionately fall on Germany, whereas the burden of taxes on portable luxury goods will disproportionately fall on France. Hence negotiations about rate differentiation will have to some extent a zero-sum character. Having a uniform rate could be a reasonable focal point. Moreover, a uniform rate might lead to a relatively balanced distribution of tax burdens, as the estimates in section 4.2 suggest.

Adoption Both of the proposed reforms, i.e. the EU-wide luxury excise and the VAT refund abolition, require unanimity among EU member states. We propose to bundle the two reforms to improve the chances of reaching unanimity.

Alternative design options In the proposal as formulated above, the European Luxury Surcharge gets implemented as an excise. However, there might be advantages to implementing it as a hybrid between an excise applied to some categories and an additive VAT applied to the others. As discussed above, for some categories of goods the appropriate price threshold is 0 anyway. For those cases, implementation via an additive VAT is possible. Given the enforcement advantages of VAT (Pomeranz (2015)), such an implementation seems advantageous. For the categories where the appropriate price threshold is substantially above 0, the excise based implementation seems preferable, given that the VAT-based implementation would create distortions as a result of the discontinuous jump in the tax at that price threshold.

3 The case for luxury taxation to compensate for lack of progressivity in income and wealth taxation

Luxury consumption is correlated with high income. As a result, luxury taxation is progressive and thus, to the extent that the status quo tax system is insufficiently progressive, luxury taxation looks promising. However, a standard objection to this argument is that the status quo tax system is a result of historical learning and societal deliberation, trading off the benefits of progressivity with its costs. Progressive luxury taxation reduces the work incentives in a way similar to how a progressive income tax does. It also makes it less attractive for high earners to stay in or move to the country with the progressive taxation. The clearest theoretical distillation of this argument is provided by Kaplow (2006): under some technical assumptions, any redistribution achieved via a tax reform making commodity taxation progressive can be achieved more efficiently by instead equalizing tax rates across all goods and directly making the income taxation more progressive.

The Kaplowian argument stands on firm grounds when discussing unilateral national tax policy in a given country. Importantly, the Kaplowian conclusion doesn't require work disincentive effects to be the primary force limiting the optimal taxation of income. In fact, there is a more straightforward argument in the case where international mobility of high earners is the primary force limiting the degree of progressivity that is optimal from a national perspective with unilateral tax policy: progressive commodity taxation is similar to progressive income taxation in decreasing a country's attractiveness for high earners. So to maintain a given level of attractiveness, a country is best off taxing commodities at the same rate to leave consumption choices undistorted and to adjust the income tax to the highest level of progressivity compatible with maintaining the targeted level of attractiveness for the high earners.

However, when it comes to discussing optimal EU-wide taxes, these arguments have to be qualified. In fact, precisely because the progressivity of income taxation has been eroded due to tax competition for the internationally mobile high earners, particularly in the EU (Kleven et al. (2014)), luxury good taxation can potentially play an important role in partially mitigating this lack of progressivity. It could be retorted that, again based on a Kaplowian perspective, there should instead be a progressive EU-wide income tax. However, agreeing on a common base for an income tax seems to be a daunting challenge. The complexity involved has been well recognized by the most prominent proposals for an EU wide additive tax on high incomes: Hennette et al. (2019) proposes to create a new legislative body, parallel to the European Parliament, to deliberate on such complex fiscal questions. In comparison to this, agreeing on a set of goods to be defined as "luxury" for the purposes of an EU-wide additive tax for EU own resources seems relatively straightforward.

Moreover, progressive commodity taxation has an added benefit compared to progressive income taxa-

tion: it also acts partly like a progressive tax on wealth, given that luxury spending is positively correlated with wealth and decreases the purchasing power of wealth Kaplow (2008). Thus the lack of progressivity of wealth taxation arising from the international mobility of people with high wealth (Agrawal et al. (2025)) provides a further rationale for progressive commodity taxation. Importantly, this form of quasi-wealth redistribution does not affect incentives to save or invest, in contrast to an actual tax on wealth. As a result, many of the concerns that have led to opposition to wealth taxation do not apply to luxury taxes. In addition, luxury taxation does not reduce private philanthropy, in contrast to progressive taxes on income and wealth.

Determining the optimal taxation of luxury goods based on these progressivity-grounded considerations is a challenging task beyond the scope of this report. Instead, we will derive normatively optimal tax rate under a very conservative approach that disregards the arguments grounded in progressivity just made. Our approach, explained in the next section, focuses entirely on a different rationale for luxury taxation, namely signaling and status-externality effects. This approach, as we will see, highlights also that even if the optimal progressivity of income and wealth taxation was achieved at the EU level, substantial EU-wide additive luxury taxes would remain efficiency-enhancing.

3.1 The optimal taxation of status goods

A further rationale for taxing luxury goods arises due to signaling and status externality effects. In the case of some luxury goods, there is strong evidence that a part of the motivation for purchasing the goods is derived from the information that it reveals to others (e.g. about the consumer's wealth/income, about the person's commitment in case of a gift). This information revelation still happens when the tax on the luxury goods gets increased. The extreme case is that of a "pure diamond good" as defined by Ng (1987): Ng (1987) notes that in the case of natural diamonds there is a functionally equivalent substitute, namely synthetic diamonds, which cost only a fraction. Given that people still buy natural diamonds, it must be the case that they are motivated by the signaling that the spending allows. In such a case, any increase in the tax is beneficial as it results functionally in the same outcome (information transmission, e.g. about the commitment of the person gifting the diamond) with less resources being spent on the good. In the more realistic case of an impure diamond good where the information transmission motive co-exists with the intrinsic enjoyment of the good, there exists a finite optimal tax rate.

A related alternative model posits that luxury goods generate status benefits (e.g. via signaling of wealth or income) to the consumer but that these status benefits are fixed-sum in the aggregate. Thus luxury consumption creates a classical externality that needs a Pigovian tax for correction. It turns out that for empirically plausible elasticity values, the optimal tax rate on luxury goods is similar in this status externality model as in the partial diamond good model. For reasons of tractability, we report the results for the status externality model, derived by Xie (2026).

The critical parameter for the calculation of the optimal tax on luxury goods turns out to be the answer to the following question: Per dollar spent on a luxury good, what is the number ϕ of dollars in benefits that a person derives from the status benefits that the consumption confers to her (e.g. via signaling of wealth and income)? In general, it is difficult to estimate this parameter. However, Bursztyn et al. (2018) exploits a rare opportunity to do that: in a randomized experiment, consumers were offered two types of functionally equivalent platinum credit cards, one type where this is visible and one type where it is not. The experiment implies $\phi = 1/3$.

From Xie (2026), we obtain the following simple formula for the optimal tax t_{luxury} on luxury in the (hy-

pothetical) setting of globally coordinated tax policy. This provides an important benchmark later on, when we will discuss how tax competition forces shape countries unilateral incentives to tax luxury. The optimal tax formula under global coordination is very simple and depends only on the tax $t_{\text{other goods}}$ prevailing on other goods, the prevailing markups and the status parameter ϕ which give how many dollars a person derives as status benefits per dollar spent on a luxury good:

$$1 + t_{\text{luxury}} = (1 + t_{\text{othergoods}}) \frac{1}{1-\phi} \frac{1 + \text{markup}_{\text{othergoods}}}{1 + markup_{\text{luxury}}}$$

We obtain the result $t_{\text{luxury}} = 0.42$ for the following estimates [2]:

$$t_{\text{other goods}} = 0.18, \phi = 1/3, \text{markup}_{\text{other goods}} = 0.12, \text{markup}_{\text{luxury}} = 0.4$$

Under this calibration, we thus obtain that luxury goods should be taxed at 42%. This calculation has not taken into account the other above-mentioned rationales (beyond status effects) for taxing luxury goods. Therefore, it should be considered a conservative estimate for the optimal tax on luxury goods. If we were to account for the fact that the progressivity of wealth and income taxes has been eroded due to tax competition and that luxury good consumption is correlated with high income and wealth then we would arrive at potentially much higher estimates for the optimal tax on luxury goods.

3.2 The self-interested case

Our discussion so far has focused on the optimal taxation of luxury in a hypothetical globally coordinated policy setting. Xie (2026) show that for personal luxury goods that when countries act unilaterally without coordination, they will end up with luxury taxes below the globally socially optimal rates[3]. These findings are approximately independent of the status/signaling parameter ϕ introduced above. They therefore provide a robust case for EU-wide luxury taxation.

The above-mentioned rationales for taxing luxury consumption at higher rates than other goods apply for each country individually. The population of each country benefits when status externalities (which plausibly arise primarily due to comparisons of people with a given country where they live) are taxed and thereby corrected.

However, there are three major additional effects that arise for any given country's calculation of how to set its luxury tax (on luxury purchases on its own territory) to maximize its own welfare:

1) The profits arising due to the mark-ups on the luxury goods and services partly accrue to workers and shareholders in other countries.

2) A part of the luxury consumption is done by visitors from other countries and they thus bear part of the tax burden.

3) Increases in the luxury tax causes people (domestic consumers and potential visitors) to go to another country for their luxury purchases.

[2]Concerning the markup on other goods, there is substantial uncertainty in the literature. Since we are focused on the EU, we use 0.12 based on Weche & Wambach (2021) and Ganglmair et al. (2021). The alternative estimate from Edmond et al. (2023), namely a markup of 0.22, is based on the US but more accurate in the sense that the estimates is directly for the cost-weighted markup. It is the estimate used by Xie (2026). In order to be conservative concerning the optimal luxury tax rate, we instead used the estimate 0.12.

[3]For example, Xie (2026) show that France's current standard VAT rate (20%) is almost exactly France's unilateral optimum (19%). For Italy (27%), Germany (31%) and Spain (36%), the unilateral optimum is above the status quo but still much below the global optimum.

The effects 1) and 2) create national incentives to set the luxury tax high. If effect 3), which pushes instead towards low tax rates was sufficiently weak then we would get the theoretical prediction that national governments have an overall incentive to set their luxury taxes above the globally optimal level. However,Xie (2026) provides an empirically calibrated model (for portable personal luxury goods) that implies that 3) is actually stronger than 1) and 2) combined. Thus we obtain the conclusion that tax competition distorts national luxury taxes downwards. This provides a strong rationale for a luxury tax at the EU level in order to correct for the forces of tax competition. Specifically, the modeling from Xie (2026) yields that the marginal cost of raising a EUR via an EU-wide luxury tax, starting from the situation where the VAT refund has been abolished, is 0.88 EUR[4]. This estimate implies that it is 12 percent cheaper for the EU to raise funding via luxury taxation than via GNI-based contributions.

Importantly, this estimate that a marginal euro of revenue raised via an EU-wide luxury tax only costs the EU countries 0.88 euros already accounts for some of the main effects that are commonly (and justifiably) adduced as arguments against such taxes in Europe. Firstly, the underlying modeling captures the effects on the EU's luxury industry. The model captures this via a markup that accrues to the luxury producers that is higher than in other sectors due to the strong brand-based differentiation characteristic of the industry. Secondly, the model takes into account cross-border shopping diversion to countries such as the UK, Switzerland and the Gulf countries and the fact that such diversion reduces the EU countries' revenue from the VAT collected on the non-refundable consumption by the luxury shoppers.

3.3 Effects on global welfare

Some luxury goods such as jewelery have large negative externalities due to the role that precious stones play in financing conflicts Binzel et al. (2023) and in undermining democratic accountability Bhattacharyya & Hodler (2010), as the large literature on the "resource curse" has documented (Olsson (2007), Blair et al. (2021)). This adds a further reason for why the optimal tax rates on the relevant luxury goods can be expected to be larger than the one calculated (based on signaling/ status externalities) above.

4 Revenue estimates

4.1 Aggregate revenue estimate

Our proposed excise involves category-specific price thresholds, with a uniform ad valorem rate being applied to the value exceeding the corresponding threshold. Defining these thresholds for each category is beyond the scope of this report. Instead, we provide a rough estimate for the revenue potential of luxury taxes assuming that the thresholds are set at 0 but that the product categories are defined in a way corresponding to luxury product status in our data sources. Assuming that the threshold gets in practice set at a relatively low level, the results presented here should give a good approximation.

For each category, let us assume that pre-tax prices are unaffected by the luxury taxes. [5] Consider the case of a luxury good (e.g. luxury cars) that is bought only by residents. Let us assume that the demand has constant price elasticity denoted by ϵ. Denoting the pre-tax price of the good by p, the country i's own ad

[4]It must be emphasized that there is large uncertainty associated with this estimate. We have not found any other estimate in the literature. The 0.88 estimate based on the model Xie (2026) is for a specific category of luxury (namely luxury leather goods) as a proof-of concept. The calibration can be straightforwardly extended to other categories with substantial additional data work.

[5]This happens in monopolistic competition models, used e.g. by Xie (2026) for the analysis of luxury good taxation.

valorem [6] tax on luxury goods by t_i and the additive EU-wide tax by t_{EU}. Then we can write the quantity consumed in terms of the status quo value, denoted Q_i^0:

$$Q_i = Q_i^0 \left(\frac{1 + t_i + t_{EU}}{1 + t_i} \right)^{\epsilon}$$

Let $T_{i,EU}$ denote the revenue from the EU-wide tax collected by country i. We get:

$$T_{i,EU} = t_{EU} p Q_i = t_{EU} p Q_i^0 \left(\frac{1 + t_i + t_{EU}}{1 + t_i} \right)^{\epsilon}$$

Now denoting by R_i^0 the status quo spending on the luxury good category by country i, we get $R_i^0 = pQ_i^0(1 + t_i)$. Given our assumption that both the pre-tax price p and the country's own tax rate are unaffected by the EU-wide additive ad valorem tax, we can thus write:

$$T_{i,EU} = t_{EU} \frac{R_i^0}{1 + t_i} \left(\frac{1 + t_i + t_{EU}}{1 + t_i} \right)^{\epsilon}$$

Rearranging gives:

$$T_{i,EU} = \frac{t_{EU}}{1 + t_i + t_{EU}} R_i^0 \left(\frac{1 + t_i + t_{EU}}{1 + t_i} \right)^{1+\epsilon}$$

Elasticity calibration For portable personal luxury goods, arguably the most directly relevant tax-base elasticity comes from the UK Office for Budget Responsibility's review of the abolition of the VAT Retail Export Scheme (OBR, 2024). The OBR's original 2020 costing assumed an elasticity of -1.9, itself a 50% scale-up of a UK tourism elasticity of -1.28 intended to capture the greater price-sensitivity of tax-free luxury shoppers; on review the OBR judged this too high and revised its central estimate to approximately -1.2. It is important to read this figure for what it measures: a *tax-base* elasticity that bundles the pure consumption response together with the diversion of tourist purchases to competing destinations—above all to Paris and Milan, which retained tax-free shopping when the UK withdrew it. The estimate is therefore dominated by a *destination-substitution* margin that is specific to a unilateral national measure.

For an EU-wide luxury surcharge the relevant elasticity is likely to be substantially smaller in absolute value. The -1.2 estimate is large precisely because a non-EU visitor deciding where to buy a handbag could avoid the UK price increase by purchasing in another European capital at no cost to their itinerary; the tax fell on one destination while its closest substitutes remained untaxed. A harmonised surcharge applied simultaneously across all 27 member states closes exactly this margin: intra-EU relocation of purchases no longer avoids the tax, and only the two weaker channels remain—genuine reductions in consumption, and diversion to destinations outside the EU (principally the UK, Switzerland, and the Gulf and Asian luxury hubs). Since a large share of the leakage embodied in the OBR figure is intra-European, the own-price elasticity of the EU-wide tax base should be expected to lie well below 1.2 in magnitude, closer to the underlying consumption elasticity of personal luxury goods, with only a modest additional wedge for extra-EU diversion.

We can also get some theoretical prior on ϵ by considering "pure diamond goods", introduced by Ng (1987)) as explained above. For pure diamond goods we can theoretically conclude that $\epsilon = -1$. For the realistic case where consumers are also motivated in part by the intrinsic enjoyment of the good, the

[6] This includes both VAT and excises if they exist.

elasticity can be below or above -1. The econometric estimates find demand to be inelastic, i.e. less elastic than 1 in absolute value:

Table 1.2: Category-level own-price elasticities of demand, luxury-relevant categories

Category	Category-level price elasticity of demand	Source
Jewellery & watches (US, PCE aggregate, 1947–2004)	-0.38 short run; -0.17 steady state	Taylor & Houthakker (2010, Table 15.15)
Gold imports, jewellery-dominated (India, aggregate)	≈ -0.5 to -1.0 long run; more elastic in the short run[a]	Kanjilal & Ghosh (2014)
Wine (aggregate consumption)	-0.69 (mean of reported estimates)	Wagenaar et al. (2009)
Spirits (aggregate consumption)	-0.80 (mean of reported estimates)	Wagenaar et al. (2009)
New passenger vehicles (US, whole market)	-0.34 (long run)	Leard (2022)

Notes: Only estimates at whole-category aggregation are included, as inter-brand substitution inflates $|\varepsilon|$ relative to the response to a category-wide tax.

A revenue estimate The results that we present here are very insensitive to the exact elasticity that we assume among the range of empirical estimates provided above, at least for the moderate tax rate of 20% . For simplicity, we therefore assume that $\epsilon = -1$ for each of the luxury categories. We get from the formula derived above the tax revenue from the EU-wide tax as a function of the status quo consumer spending R_i^0, the status quo national tax rate t_i and the EU-wide tax rate t_{EU}, :

$$T_{i,EU} = R_i^0 \frac{t_{EU}}{1 + t_i + t_{EU}}$$

We see that the amount of tax revenue that the EU-wide tax rate generates increases in the tax rate t_{EU} and approaches 299 billion euro as the tax rate goes to infinity.

Let us now compute the tax revenue for the case where all EU countries have a tax rate on luxury goods equal to the average statutory rate, i.e. 22%:

$$\sum_{i \in EU} T_{i,EU} = \sum_{i \in EU} R_i^0 \frac{t_{EU}}{1 + 0.22 + t_{EU}} = 299 \cdot \frac{t_{EU}}{1.22 + t_{EU}}$$

Consider the case where we set $t_{EU} = 0.2$ so that the overall ad valorem tax rate is equal to 42%, the (conservatively estimated) optimal rate according to Xie (2026). Then we get for the aggregate revenue collected via the EU-wide luxury tax:

$$\sum_{i \in EU} T_{i,EU} = 299 \cdot \frac{0.2}{1.42} = \text{EUR 42 billion.}$$

Further revenue potential The revenue estimate of 42 billion euros that we just provided is conservative in at least two ways: Firstly, the (surprisingly weak) empirical evidence surveyed above is consistent with

the demand actually being substantially less elastic than our assumed number of $\epsilon = -1$. This would imply that the maximal theoretically achievable *revenue potential* is larger than the 299 billion euros that we have for $\epsilon = -1$.

Of course, the socially optimal tax on luxury goods is likely much below the rate that maximizes tax revenue. At some point, excessively high luxury tax rates could reduce work incentives for those of the high-earners that are motivated by the intrinsic enjoyment that they derive from consuming luxury goods. However, given that the tax progressivity in the EU has been pushed down by tax competition for people with high income and wealth, the optimal tax on luxury goods is likely substantially above the 42% that is justified by status externalities alone. In fact, in the theoretical scenario where the price elasticity of overall luxury demand is close to 0, the luxury tax can make up for most of the lack of progressivity of the income and wealth taxation in the EU.

Quantifying the overall optimal luxury tax is beyond the scope of this report. However, let us illustrate the revenue potential under the above assumptions:

For a 40% tax rate, tax revenue would be EUR 74 billion.

For a 60% tax rate, tax revenue would be EUR 99 billion.

4.2 Revenue collected by Member State

As motivated above, this report proposes the use of a uniform EU-wide tax rate across all luxury goods to fund the EU budget as an own resource. One important design choice implicit in this proposal is to bundle all the luxury tax bases together, rather than putting forward luxury taxes on specific categories as separate proposals. The following estimates provide one of the motivations for such a broadly bundled approach by showing that the tax revenue collected and thus contributed to the EU budget is not that far away to being proportional to GDP.

This purely descriptive finding suggests that the proposal has some promise for garnering unanimity if countries are well-informed about their self-interest, under two reasonable assumptions. Firstly, let us assume that the benefits from the EU spending is roughly proportional to GDP. Secondly, let us assume that the tax burden is roughly proportional to the tax revenue collected. This latter assumption is not necessarily accurate as there are other components in the overall tax burden, most notably profits and international shifting of consumers. However, the proportionality assumption likely provides a first decent approximation for the distribution of tax burdens across countries.

Under these two assumptions, it follows that at a given overall benefit-cost ratio of the proposed reform (additive EU-wide luxury tax to fund EU budget) is more likely to garner unanimity if the ratios of tax revenue to GDP are less dispersed across countries. To see why, note that the impact of additional own resources is to either expand EU spending or to replace the residual contributions that are proportional GNI which itself is close to GDP. If all countries had the same ratio of tax revenue collected to GDP then rational voting behaviour would yield unanimity as long as the aggregate benefit-cost ratio exceeds 1. This is highly plausible, as we will argue below.

We use the definition of a luxury good/service via the list of categories provided above (Automotive, Personal luxury goods, Hospitality (hotels, cruises), Yachts, Fine wines and spirits, Gourmet food and fine dining, Design and furniture). Thus the following estimates are the aggregates across all these categories.

We see that, setting aside the small countries Cyprus and Malta[7], the maximal ratio of a country's implied share of luxury tax revenue contributed to the EU to its share of EU GDP is 1.61 (see Greece). Thus

[7] where maybe correspondingly small ad hoc transfers could be appropriate to compensate them

Table 1.3: EU-27 luxury tax base by member state, EUR bn per year, net of VAT.

Member state	Base EUR bn	Rev. %	GDP %	Ratio	Member state	Base EUR bn	Rev. %	GDP %	Ratio
France	62.0	20.7	16.3	1.27	Romania	3.2	1.1	2.2	0.49
Germany	57.7	19.3	24.5	0.79	Finland	2.9	1.0	1.6	0.61
Italy	56.5	18.9	12.3	1.54	Hungary	2.5	0.9	1.1	0.78
Spain	30.9	10.3	8.7	1.19	Croatia	2.3	0.8	0.5	1.55
Netherlands	11.0	3.7	6.0	0.61	Luxembourg	1.8	0.6	0.5	1.21
Austria	9.9	3.3	2.8	1.18	Bulgaria	1.6	0.5	0.6	0.88
Belgium	9.8	3.3	3.5	0.93	Cyprus	1.5	0.5	0.2	2.44
Poland	8.0	2.7	4.7	0.57	Slovakia	1.3	0.4	0.7	0.63
Sweden	7.1	2.4	3.0	0.79	Slovenia	1.1	0.4	0.4	0.93
Portugal	6.4	2.1	1.6	1.34	Malta	0.9	0.3	0.1	3.11
Greece	6.3	2.1	1.3	1.61	Lithuania	0.8	0.3	0.5	0.53
Czechia	4.6	1.5	2.0	0.77	Estonia	0.5	0.2	0.2	0.91
Denmark	4.2	1.4	2.3	0.62	Latvia	0.5	0.2	0.3	0.55
Ireland	3.8	1.3	3.0	0.43	**EU-27**	**299**	**100**	**100**	

"Rev. %" is the share of total revenue collected; "GDP %" the member state's share of EU-27 GDP in 2024; "Ratio" the former divided by the latter. The estimates apply to the case of a small tax, where tax revenues are proportional to the status quo tax base. The estimates use data from Eurostat(2024), ACEA (2026), ECCIA–Bainreport (2025) and ad hoc assumptions that would take too much space to detail here. The method is explained in the methodological note and implemented in the accompanying code.

under rational voting behaviour (and the above assumption) unanimity holds iff 1 euro of spending by the EU yields at least 1.61 in benefits for EU countries overall.

In reality, the probability of unanimity is of course not just determined by the largest relative tax burden given that countries at somewhat lower relative tax burdens (e.g. Croatia) could also oppose the reform. Thus, broadly it is favorable for unanimity to have a low dispersion in the ratio of tax revenue share to GDP share. It turns out that this dispersion is 36% lower than the average dispersion that would result if we considered each of the sectors separately.[8] The reason for this is that the sectors are far from being perfectly correlated. For example Germany bears a disproportionate part of the burden from luxury car taxation and a relatively small part of the burden from personal luxury good taxation, whereas the reverse is true for France.

4.3 Should countries with high relative tax burden be compensated?

A natural approach for compensating countries with high relative tax burden could be to reduce their required GNI-based contributions. If countries believe that EU own resources substitute 1 for 1 for GNI-based contributions then this could be seen as favorable for facilitating unanimity. However, there are substantial downsides associated with such an approach arising from the fact that it would aggravate countries' tax competition for people with high wealth and income. To see why, consider the extreme case where the GNI-based contribution is fully adjusted so that each country ends up making a net contribution to the EU budget corresponding to its GNI share. In this case, a country de facto gets to fully keep every euro of additional luxury tax revenue, given that it gets compensated one for one for exceeding the tax revenue corresponding to its GNI share. The country would therefore have an incentive to lower its taxes on high

[8] A standard way of measuring the dispersion is via the GDP-weighted coefficient of variation of the ratio. For this we get: A base-weighted average of the sector variances would imply an aggregate CV of 0.557; the actual aggregate CV is 0.354. Roughly 36% of the dispersion cancels when the sectors are combined.

income and high wealth due to the increased gain from attracting such people, given that their disproportionate spending on luxury goods.

The detrimental effect of compensating countries for high luxury tax contributions to the EU with reduced GNI-based contributions that we just highlighted scales with the size of the corresponding country's luxury tax base. A pragmatic approach is therefore to resort to this form of compensation for all countries who are sufficiently small in this sense and who are sufficiently disproportionately burdened by the luxury tax. Such an approach can increase the chances of reaching unanimity without compromising too much on the welfare gains that the reform can achieve.

5 The case for using the luxury tax as EU own resource

Tax competition between EU member states undermines the unilateral incentives for EU countries to introduce and raise luxury taxes. This international spillover suggests that there might be aggregate gains for EU member states to jointly increase luxury good taxes even if it is not in the individual interest of any country to do this individually. This motivates the proposal to introducing a new EU-wide ad valorem luxury excise. Empirically calibrated modeling (Xie (2026)) suggests that there are already some countries who would benefit from unilaterally introducing luxury taxes. However, the collective economic gains from luxury taxes are much larger.

The introduction of such a new additive EU-wide luxury taxrequires unanimity between member states. For this it is, other things equal, favorable if the revenue raised creates large collective benefits. Thus it could be the case that using the tax revenue for the EU budget as an own resource, as we propose, could actually make it more likely that all member states vote for it.

5.1 Bundling to facilitate unanimity

The proposed European Luxury Surcharge as an EU own resource purports to simultaneously address two current inefficiencies: 1) suboptimal luxury good tax rates driven by tax competition and 2) suboptimal funding for the EU (in particular EU-wide international public goods).

Alternatively, one could have two separate EU-wide policy reforms, each addressing one of these problems: Firstly, one could have a scheme where an EU-wide luxury tax gets introduced and each member state gets to keep its revenue. Secondly, one could have a scheme where each member state has to contribute additional funding for the EU, e.g. in proportion to its GNI as in the traditional EU funding scheme. One could call this alternative the "separated approach", with a separate vote for each of them at the EU level (each requiring unanimity).

This separation approach has been the dominant approach, both globally and at the EU level. For example, the Global Minimum Tax on corporate income does not impose any earmarking conditions. Instead, it focuses on addressing a race-to-the bottom in corporate income tax rates and, relatedly, the inefficiency of transfer-pricing based tax avoidance. Funding of global funds for global publics goods and redistribution is generated separately from any tax coordination. It is based on voluntary contributions and, in the case of the UN institutions, partly on assessed contributions calculated in proportion to GDP that countries have to pay to keep their voting rights at the UN general assembly.

Similarly, at the EU level, the separation approach has been the dominant one. Most existing mechanisms for tax coordination impose rules on member states without any earmarking obligations. For exam-

ple, there are restrictions on the set of VAT rates that member states are allowed to impose. Mostly, tax revenues are retained by member states. Small exceptions include the EU-wide plastic own resource that both addresses an international externality and contributes to the EU budget. However, most of the funding for the EU budget comes from GNI-based contributions and tax-based contributions (e.g. the required 0.3% of VAT tax base) do not (aim to) address tax competition or other under-taxation issues.

In the case of the EU, the separation approach might be motivated as follows. If the international public goods that the EU budget in part provides generate benefits that are roughly proportional to countries' GNI then funding them via GNI-based contributions (or VAT base contributions that themselves are roughly proportional to GNI) ensures that all countries are better off as a result, as long as the international public goods yield more than 1 EUR in aggregate benefits per 1 EUR spent on them, a condition synonymous with under-provision, which clearly is the case we are in.

However, such exact proportionality of the GNIs and the benefits that countries derive from additional spending by the EU is unlikely to hold in practice. This could explain the difficulties of getting agreement on expanding GNI-based contributions. Similarly, there might be some countries that will not gain from EU-wide luxury taxes if the revenue is retained nationally. However, if the opposing countries differ between the two components (increasing EU spending and increasing EU-wide luxury taxation), then the bundle might still garner unanimity.

5.2 Bundling to avoid aggravating other tax competition

A further argument in favor of the bundled approach exemplified by EU own resource design is that it can alleviate tax competition on other (potentially hard to contract on) tax instruments, in contrast to the separated approach. Specifically, consider the scenario where (under a separated approach) a minimum tax on luxury goods has been adopted at the EU level, where members states are obliged to impose it but can fully retain the tax revenue from it. Such a scheme would increase the value for each government of attracting top earners that buy luxury goods as fiscal residents, given that now the government will collect the luxury tax on this. As a result, the tax competition forces that depress top income tax rates in the EU (Muñoz (2021), Kleven et al. (2014)) would get intensified.

This is in stark contrast to the case of the EU own resource. In that case, the new luxury tax revenue would not generate a strong additional incentive for EU countries to attract high earners. To see why, note firstly that each government in the EU will value a euro collected in revenue from the common luxury tax at typically much less than 1 euro, given that it cannot retain it and given that only a small part of the international public good benefits generated by the EU spending accrue to any given country. Moreover, if high earners move to other EU countries, they will also generate this luxury tax revenue for the EU budget, further reducing a given government concern for attracting high earners from other EU countries. As a result, the EU own resource design would not substantially aggravate top income tax competition within the EU, in contrast to the separated approach.

6 Domestic compensation

Throughout this report, we have focused on each country's economic interest at an aggregate level. We have argued that the proposed reforms could make each EU member state better off. However, this does not preclude the possibility that the reforms might make some groups of workers in specific luxury sectors

worse off. Therefore, it is important to combine the proposed reforms with domestic compensation schemes to ensure that ideally an overwhelming majority of the population in each EU member state would gain.

Three groups bear the incidence of luxury taxes. Luxury consumers, shareholders of luxury firms and luxury sector workers. For the first two groups, compensation seems unnecessary. Luxury consumers actually reap some of the welfare gains of luxury taxes in the form of reduced resources expended on status competition. On net, they still bear a burden from the tax. In the case of portable luxury goods consumers, the results from Xie (2026) suggest that consumers bear most of the net incidence. However, given that they have typically high income and wealth, there is no strong rationale for compensating them. Similarly, it seems unnecessary to compensate shareholders of luxury firms.

Concerning the workers in the luxury sector, however, compensation seems very important for building support for the proposed reforms. As the luxury tax will lower production of the luxury products, the design of appropriate compensation policies must take into account how difficult it is for workers to transition to other segments or sectors. In some sectors, transitioning from the luxury segment to a non-luxury segment (or another sector) might have low costs. This might be the case for hospitality. In this case, the incidence of a luxury tax will be born by all workers in the hospitality sector. Reducing the indirect taxes on the hospitality sector could then precisely offset the incidence[9].

In sectors where workers' skills strongly depreciate as they move away from their specialized work in the luxury sector (e.g. in the luxury car industry), compensation policies could take the form of specific wage subsidies to workers displaced from the luxury sector.

Such compensation policies could be funded by dedicating some fraction of the luxury tax revenue to them. This could be achieved by letting EU member states retain some fraction of the EU-wide luxury surcharge. Alternatively, under the design of our baseline proposal for the EU-wide luxury surcharge, where all the revenue would be directed to the EU, EU member states could implement additive national rates. Indeed, under the excise-based implementation there would be no legal obstacles for governments to introduce such additive rates. Moreover, they could even make them sector-specific, reflecting their own preferences and interests. The results from Xie (2026) for personal luxury goods suggest that governments would have incentives to do so[10].

[9] reduced VAT rates on (non-luxury) accommodation and restaurant services are already available to them under Annex III of the VAT Directive, with additional rate flexibility since the 2022 reform of the EU rate rules through Council Directive (EU) 2022/542 of 5 April 2022. A further loosening of the restrictions on the number and level of reduced rates may in any case come onto the agenda if the European Luxury Surcharge is partly implemented as an additive VAT.

[10] They also raise the question as to why governments do not already impose such excises for themselves. One reason could be the fixed (administrative) costs of introducing such an excise are perceived to outweigh the gains. The introduction of the EU-wide luxury surcharge would mean that there would be no further administrative costs of introducing an additive nationally retained rate. Moreover, the earmarking of the EU-wide luxury surcharge would preserve countries' unilateral incentives to add national excises, given that the tax competition would be dampened (Keen & Kotsogiannis (2002), Devereux et al. (2007), Besley & Rosen (1998)).

Bibliography

D. R. Agrawal, D. Foremny, & C. Martínez-Toledano. Wealth tax mobility and tax coordination. *American Economic Journal: Applied Economics*, 2025. https://doi.org/10.1257/app.20220615. 7

T. J. Besley & H. S. Rosen. Vertical externalities in tax setting: Evidence from gasoline and cigarettes. *Journal of Public Economics*, 1998. https://doi.org/10.1016/S0047-2727(98)00041-3. 16

S. Bhattacharyya & R. Hodler. Natural resources, democracy and corruption. *European Economic Review*, 2010. https://doi.org/10.1016/j.euroecorev.2009.10.004. 9

C. Binzel, D. Fehr, & A. Link. Can international initiatives promote peace? Diamond certification and armed conflicts in Africa. CEPR Discussion Paper 18450, Centre for Economic Policy Research, Paris and London, 2023. https://cepr.org/publications/dp18450. 9

G. Blair, D. Christensen, & A. Rudkin. Do commodity price shocks cause armed conflict? A meta-analysis of natural experiments. *American Political Science Review*, 2021. https://doi.org/10.1017/S0003055420000957. 9

Bloomberg & J. Creery. Post-Brexit shoppers shift from London to tax-free Paris and Milan in 'lost opportunity'. *Fortune*, 2024. https://fortune.com/europe/2024/05/09/shoppers-tax-free-london-paris-milan-post-brexit/. 3

L. Bursztyn, B. Ferman, S. Fiorin, M. Kanz, & G. Rao. Status goods: Experimental evidence from platinum credit cards. *The Quarterly Journal of Economics*, 2018. https://doi.org/10.1093/qje/qjx048. 7

C. D'Arpizio, F. Levato, A. Steiner, & J. de Montgolfier. Altagamma–bain luxury goods worldwide market study, 23rd edition: Good times for a change. Technical report, Bain & Company and Fondazione Altagamma, 2024. https://altagamma.it/media/source/Altagamma-Bain%20WW%20Luxury%20Market%20Monitor%202024.pdf. 4

DAT. DAT-report 2026 – kurzbericht. Technical report, Deutsche Automobil Treuhand GmbH, Ostfildern, 2026. https://www.dat.de/fileadmin/de/images/produkte/DAT_Report/DAT_Report_NEW/2026/DAT-Report-2026-Kurzbericht.pdf. 4

M. P. Devereux, B. Lockwood, & M. Redoano. Horizontal and vertical indirect tax competition: Theory and some evidence from the USA. *Journal of Public Economics*, 2007. https://doi.org/10.1016/j.jpubeco.2006.07.005. 16

C. Edmond, V. Midrigan, & D. Y. Xu. How costly are markups? *Journal of Political Economy*, 2023. https://doi.org/10.1086/722986. 8

European Cultural and Creative Industries Alliance & Bain & Company. The contribution of the high-end cultural and creative sectors to the european economy. Technical report, European Cultural and Creative Industries Alliance, 2025. https://www.eccia.eu/assets/activities/files/ECCIA%20Bain%20Report_Walpole%20FINAL%20Digital%20Singles.pdf. 4

B. Ganglmair, A. Kann, & I. Tsanko. Markups for consumers. *Jahrbücher für Nationalökonomie und Statistik (Journal of Economics and Statistics)*, 2021. https://doi.org/10.1515/jbnst-2020-0064. 8

S. Hennette, T. Piketty, G. Sacriste, & A. Vauchez. *How to democratize Europe*. Harvard University Press, 2019. https://doi.org/10.4159/9780674239616. 6

K. Kanjilal & S. Ghosh. Income and price elasticity of gold import demand in India: Empirical evidence from threshold and ARDL bounds test cointegration. *Resources Policy*, 2014. https://doi.org/10.1016/j.resourpol.2014.05.003. 11

L. Kaplow. On the undesirability of commodity taxation even when income taxation is not optimal. *Journal of Public Economics*, 2006. https://doi.org/10.1016/j.jpubeco.2005.07.001. 6

L. Kaplow. *The Theory of Taxation and Public Economics*. Princeton University Press, Princeton, NJ, 2008. 7

M. J. Keen & C. Kotsogiannis. Does federalism lead to excessively high taxes? *American Economic Review*, 2002. https://doi.org/10.1257/000282802760015784. 16

H. J. Kleven, C. Landais, E. Saez, & E. Schultz. Migration and wage effects of taxing top earners: Evidence from the foreigners' tax scheme in Denmark. *The Quarterly Journal of Economics*, 2014. https://doi.org/10.1093/qje/qjt033. 6, 15

B. Leard. Estimating consumer substitution between new and used passenger vehicles. *Journal of the Association of Environmental and Resource Economists*, 2022. https://doi.org/10.1086/715814. 11

M. Muñoz. Do European top earners react to labour taxation through migration? PSE Working Paper 2021-35, Paris School of Economics, 2021. https://shs.hal.science/halshs-03252899. 15

Y.-K. Ng. Diamonds are a government's best friend: burden-free taxes on goods valued for their values. *The American Economic Review*, 1987. 7, 10

O. Olsson. Conflict diamonds. *Journal of Development Economics*, 2007. https://doi.org/10.1016/j.jdeveco.2005.07.004. 9

D. Pomeranz. No taxation without information: Deterrence and self-enforcement in the value added tax. *American Economic Review*, 2015. https://doi.org/10.1257/aer.20130393. 6

L. D. Taylor & H. S. Houthakker. *Consumer Demand in the United States: Prices, Income, and Consumption Behavior*. Springer, New York, 3rd edition, 2010. https://doi.org/10.1007/978-1-4419-0510-9. 11

UNRAE. Prezzi delle auto aumentati di 1.000 Euro nel 2024? Press release, 13 January 2025, and *Book UNRAE 2024*, 2025. https://unrae.it/sala-stampa/altri-comunicati/7005/prezzi-delle-auto-aumentati-di-1000-euro-nel-2024. 4

A. C. Wagenaar, M. J. Salois, & K. A. Komro. Effects of beverage alcohol price and tax levels on drinking: a meta-analysis of 1003 estimates from 112 studies. *Addiction*, 2009. https://doi.org/10.1111/j.1360-0443.2008.02438.x. 11

J. P. Weche & A. Wambach. The fall and rise of market power in europe. *Jahrbücher für Nationalökonomie und Statistik (Journal of Economics and Statistics)*, 2021. https://doi.org/10.1515/jbnst-2020-0047. 8

X. Xie. Federal taxation of luxury goods under internationally mobile consumers. 2026. https://drive.google.com/file/d/1LBeZ33T_Uc8QMUzgSicZy6D8HJtwO2t6/view. 7, 8, 9, 11, 14, 16